

Total No. of Questions : 8]

SEAT No. :

P229

[Total No. of Pages : 2

[5871]-756

B.E. (Honors)

INTERNET OF THINGS SECURITY
(2015 Pattern) (Semester - II) (410603)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Draw neat diagram whenever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) Draw general structure of IoT system and explain how digital signature can be used to maintain integrity and authenticity of digital contents. [6]
- b) Explain two commonly used identity authentication protocols used for IoT environment in detail. [5]
- c) What is public key cryptography? How can it help to secure IoT applications? [6]

OR

- Q2)** a) List different protocols developed for IP connectivity at different layers of IoT. Explain certificate based DTLS handshake procedure with neat diagram. [6]
- b) Explain CoAP protocol for IoT environment. [5]
- c) Briefly discuss X.509 certificate scheme. [6]

- Q3)** a) Explain the data life cycle in IoT. How can we protect data in IoT? [6]
- b) Write a short note on IoT security for machine learning applications. [6]
- c) List and explain the application-interface layer security threats in IoT. [6]

OR

- Q4)** a) State the security challenges in IoT and describe the current schemes for IoT security. [8]
- b) Draw and explain the generic SOA architecture for IoT environment. Explain the security concerns to be considered in this architecture. [10]

P.T.O.

- Q5)** a) Identify the security issues in integration of RFID and WSNs. Describe in detail. [6]
- b) How do maintain confidentiality and security for IoT based health care applications? [6]
- c) Which security schemes can be applied to protect information in Wireless Sensor Networks (WSNs)? Discuss. [6]

OR

- Q6)** a) Discuss the security attacks in WSNs. [6]
- b) List multiple systems/devices involved in identification and tracking technologies of IoT environment. Discuss threats and challenges to achieve security in these systems. [6]
- c) List and explain main potential attacks in 6LoWPAN. Discuss security protocols and privacy issues in 6LoWPAN. [6]
- Q7)** a) Draw and explain the system model for Intelligent Traffic system. [6]
- b) What is the role of IoT in connected cars? Write a short note on “Connected cars security and automation”. [6]
- c) Briefly discuss the case study on “Smart electric vehicle scheduling”. [5]

OR

- Q8)** a) Describe blockchain enabled food supply chain traceability system in detail. [6]
- b) Comment on “IoT in Smart Cities”. [5]
- c) Provide IoT-based solution for “smart home” use case. [6]
